

Gaven Jesse

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Education

Harvey Mudd College – B.S. Engineering (Distinction) May 2026
Bellevue College – A.S. in Mathematics, Physics, and Atmospheric Sciences (High Distinction) June 2022

Skills

Fabrication: Advanced metal and woodshop machinery, including CNC milling, metal lathe, and table saw. CAD (proficient)
Programming: Python & MATLAB (proficient), C & SystemVerilog (intermediate), and MotionGenesis (proficient)

Professional Experience

Sandia National Labs Materials Engineering Internship (Harvey Mudd Clinic Program) Aug 2025 – May 2026
Spring 2026 Team Lead:

- Led computational and experimental subteams to evaluate how material composition and particle size impact dielectric performance
- Synthesized experimental and simulation results to identify optimal material configurations and fabrication approaches
- Communicated technical findings to Sandia stakeholders, translating complex material behavior into clear, actionable insights
- Mentored 3 junior team members, providing technical guidance and improving onboarding and team efficiency
- R. De Santos, C. Hernandez, G. Jesse, B. Harsch, E. Hernandez, A. Dato, T. Monson, R. Van Ginhoven, “Size and loading effects on the dielectric properties of barium titanate–PDMS nanocomposites”, American Chemical Society Spring 2026 Meeting, Atlanta, GA, 3/22 – 3/26 (2026).

Fall 2025 Experimental Team Lead:

- Led fabrication and dielectric characterization of BTO–PDMS thin-film composites, establishing repeatable processing protocols
- Worked with faculty and Sandia scientists to interpret experimental results and identify likely mechanisms driving material performance
- Presented technical findings to large audiences, translating complex materials behavior into clear, accessible explanations
- Authored a 60+ page technical report documenting methods, results, and recommended next steps for continued development

Harvey Mudd College Workshop Proctor

Jan 2026 – May 2026

- Led students of varying skill levels through the fabrication of ambitious wood and metal shop projects
- Designed detailed process routers that standardized machining workflows and enabled independent execution of multi-step fabrication procedures

Harvey Mudd College Machine Shop Proctor

Fall 2025 – May 2026

- Supervised safe and effective use of metal and wood shop equipment, including mills, lathes, bandsaws, table saws, and CNC machines
- Trained and mentored students in machining practices, shop safety, and equipment-specific procedures
- Supported student projects by troubleshooting fabrication issues and guiding design for manufacturing decisions

AirMyne Mechanical Engineering Internship (Harvey Mudd Clinic Program)

August 2024 – December 2024

- Evaluated mechanical design options, balancing performance, manufacturability, and system constraints to inform final design decisions
- Presented research findings, design improvements, and working prototypes to AirMyne stakeholders; incorporated feedback into design revisions
- Coordinated team milestones and managed technical challenges while adhering to timeline and budget constraints
- Produced a midyear technical report documenting design rationale, build details, testing progress, and next-step recommendations

Research

Harvey Mudd College Energy & Nanomaterials Research

January 2024 – May 2025

- Developed and refined a repeatable fabrication process for electrically conductive composite capacitors using advanced lab techniques and materials engineering principles
- Evaluated how fabrication methods and material composition affected composite electrical performance
- Documented fabrication methods and testing procedures to improve repeatability and support future researchers
- Onboarded and mentored new team members, ensuring safe and effective experimental execution

Computer Science Startup Research

May 2023 – July 2023

- Collaborated on a 7-person startup team to design and develop Bytes, an iOS food delivery app using Python and React Native
- Presented the prototype, product strategy, and business case to large audiences and prospective investors